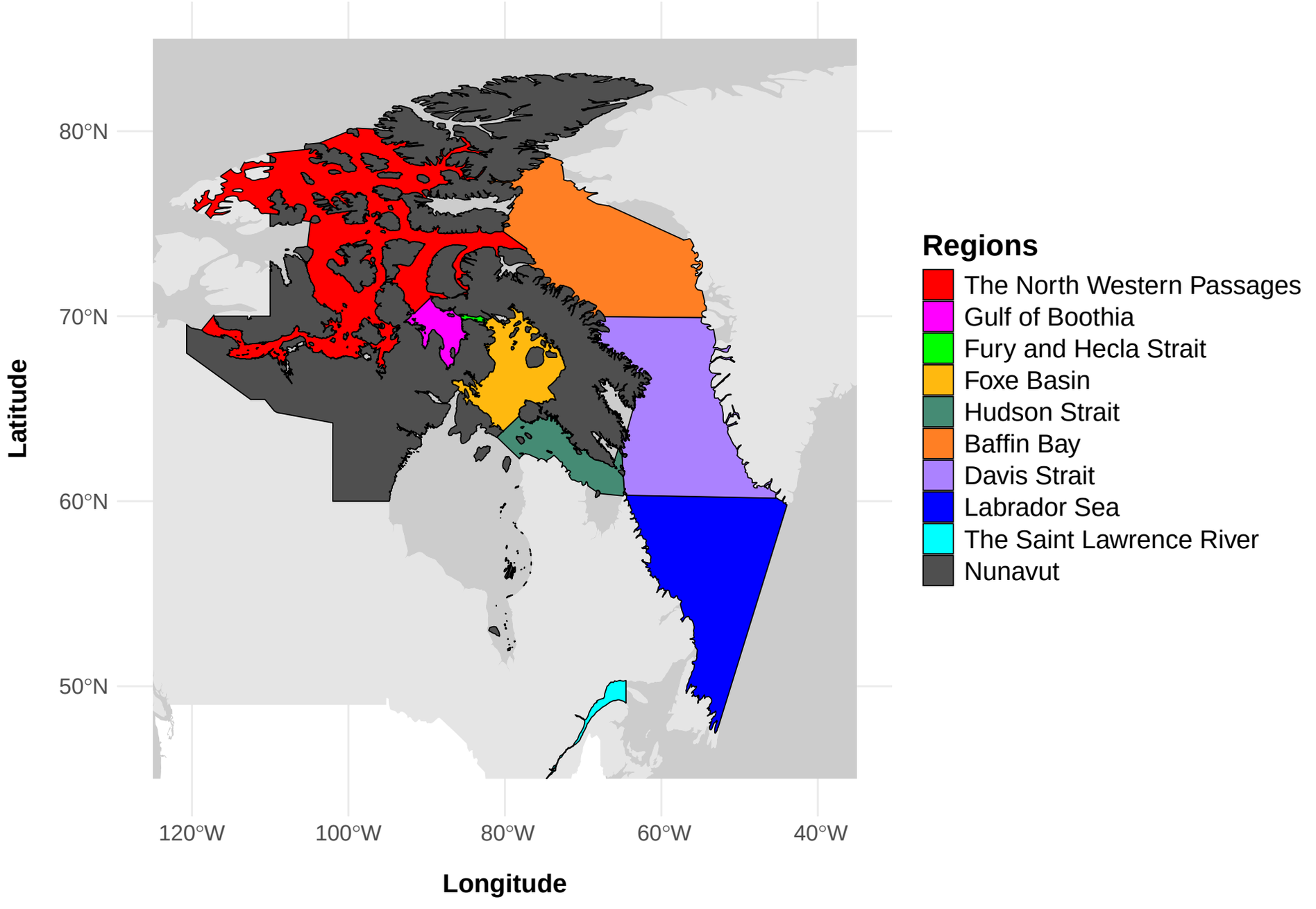
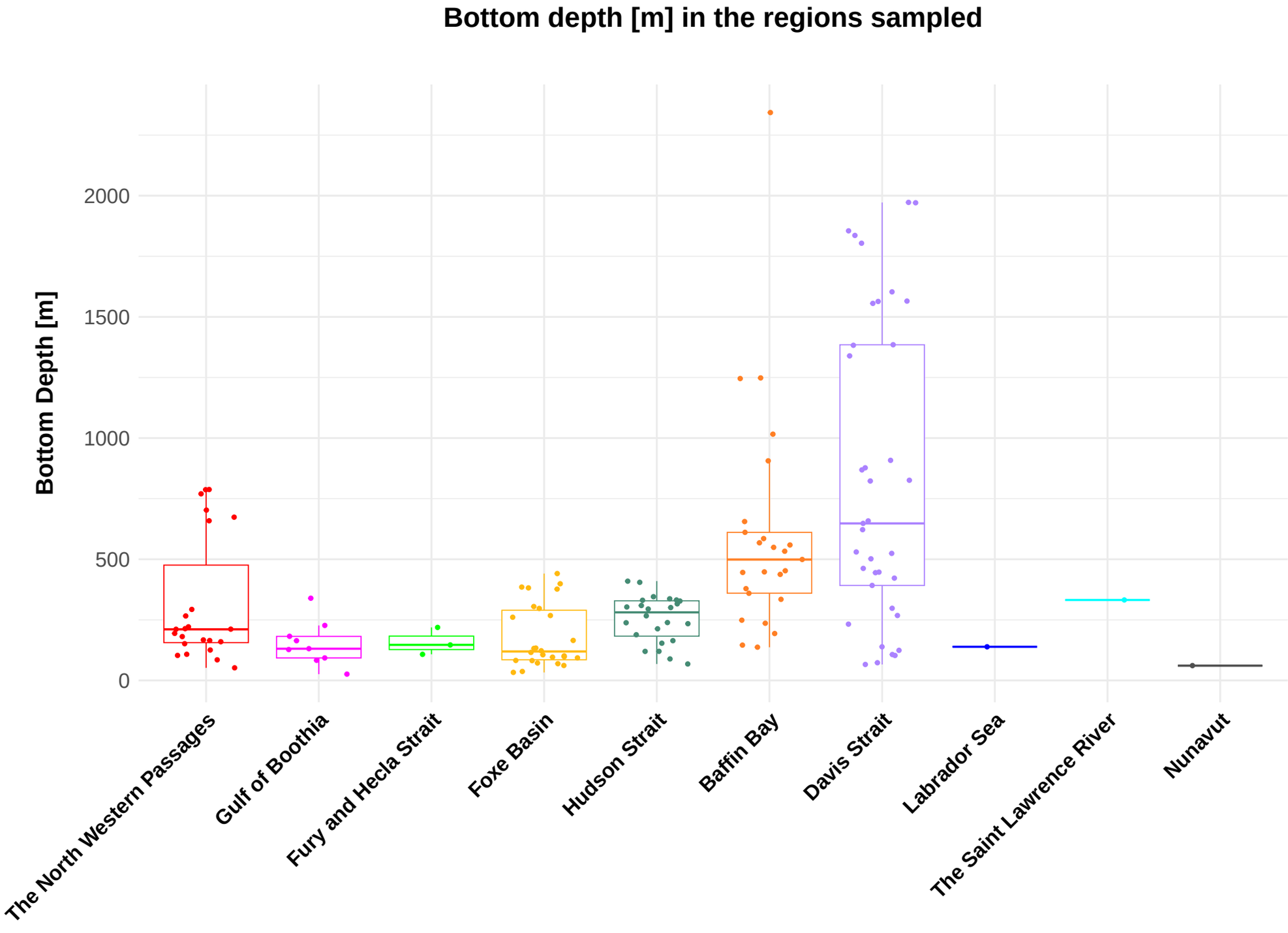
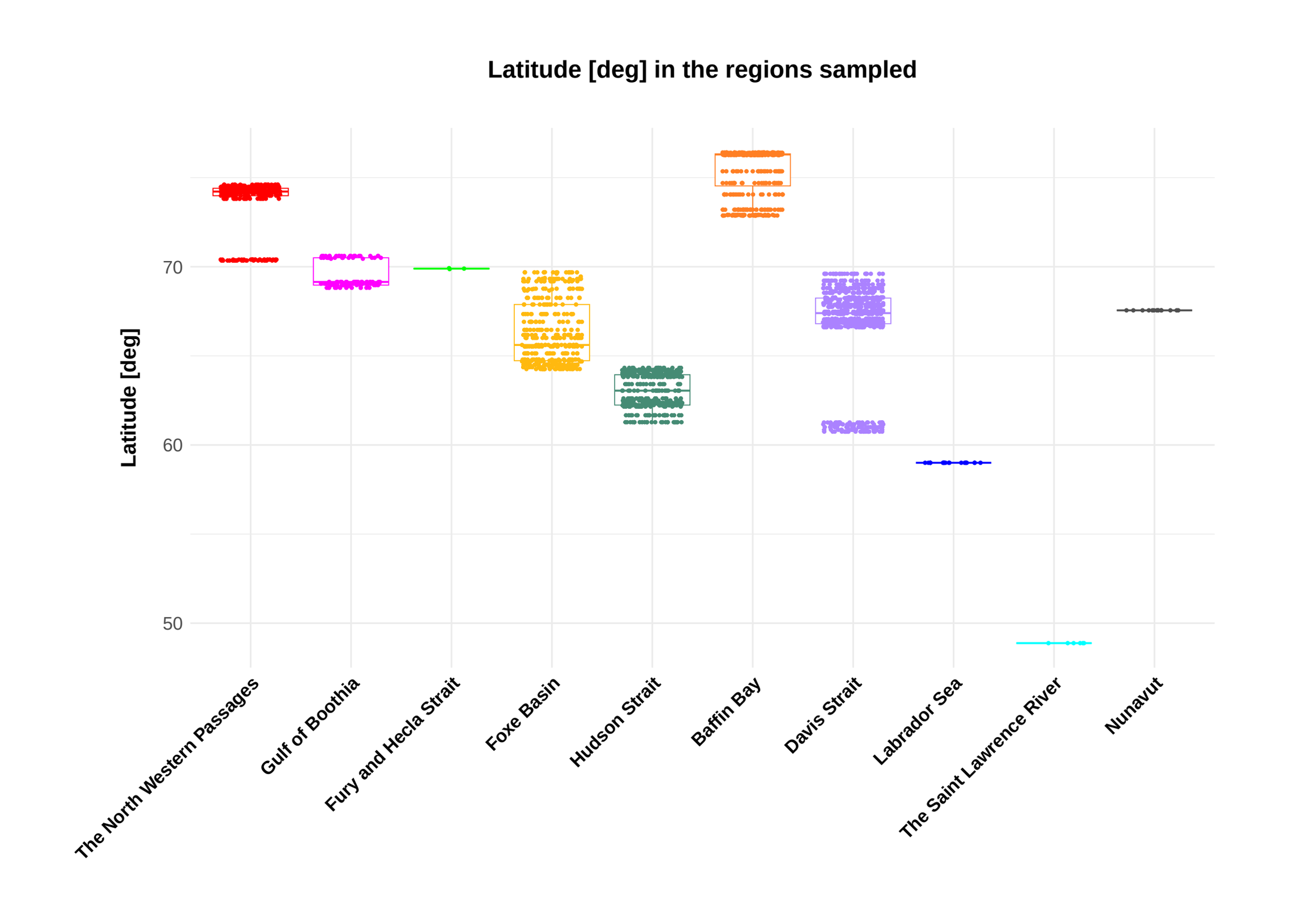
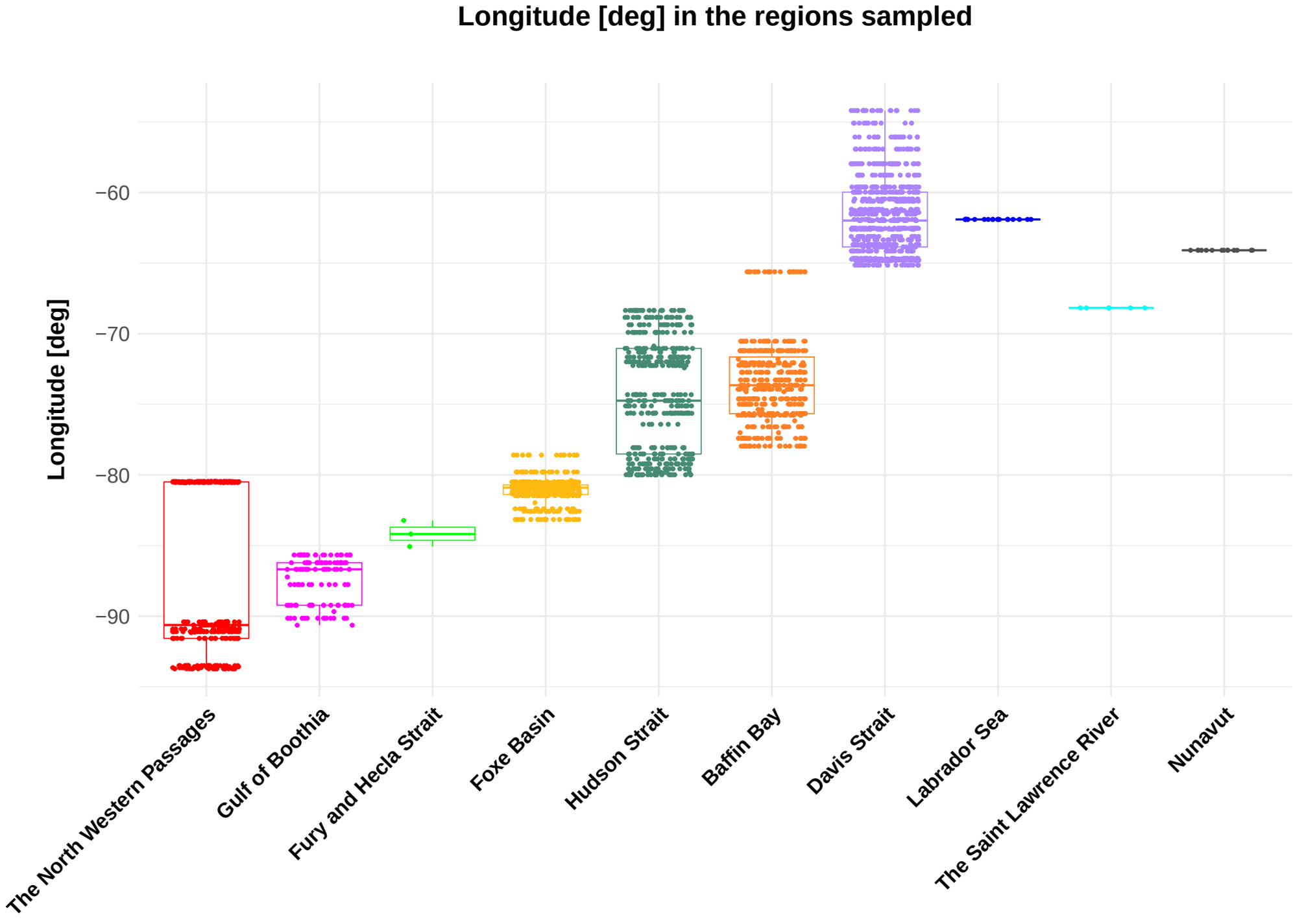
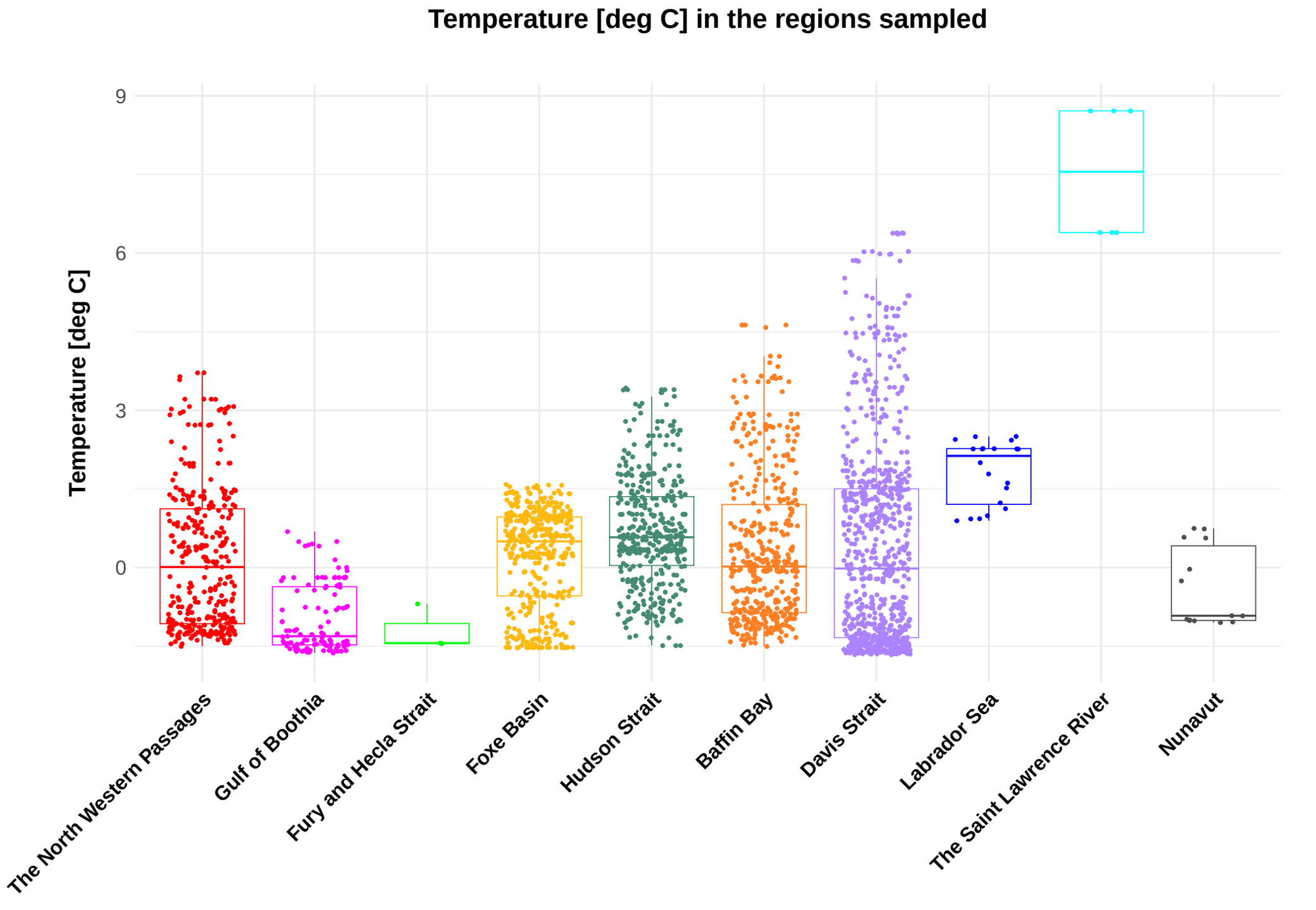
Regions sampled during Leg 2 and 4 of Amundsen 2023



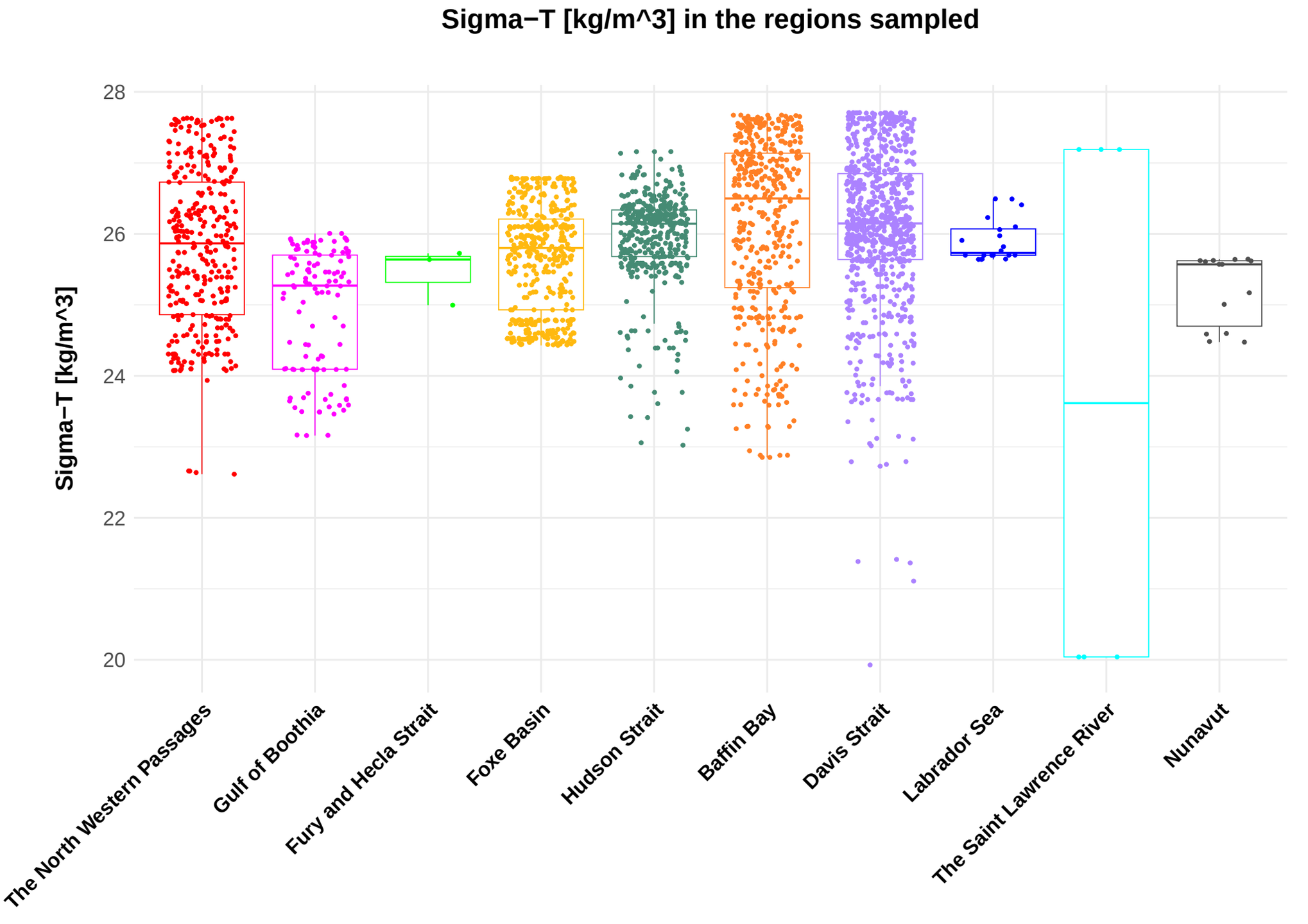

Bottom depth [m] in the regions sampled
Bottom Depth [m]
0
500
1000
1500
2000
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

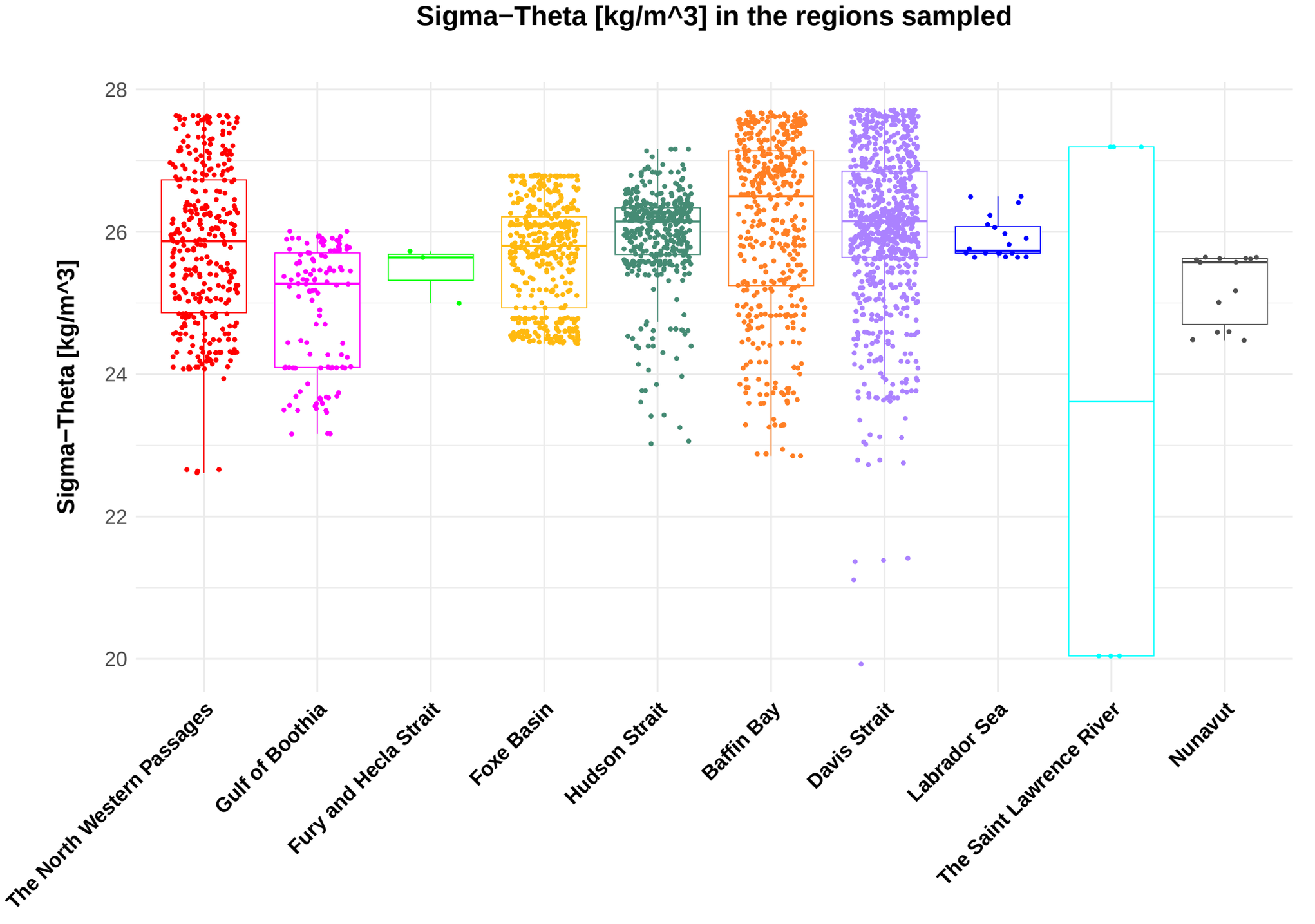

Latitude [deg] in the regions sampled
Latitude [deg]
70
60
50
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

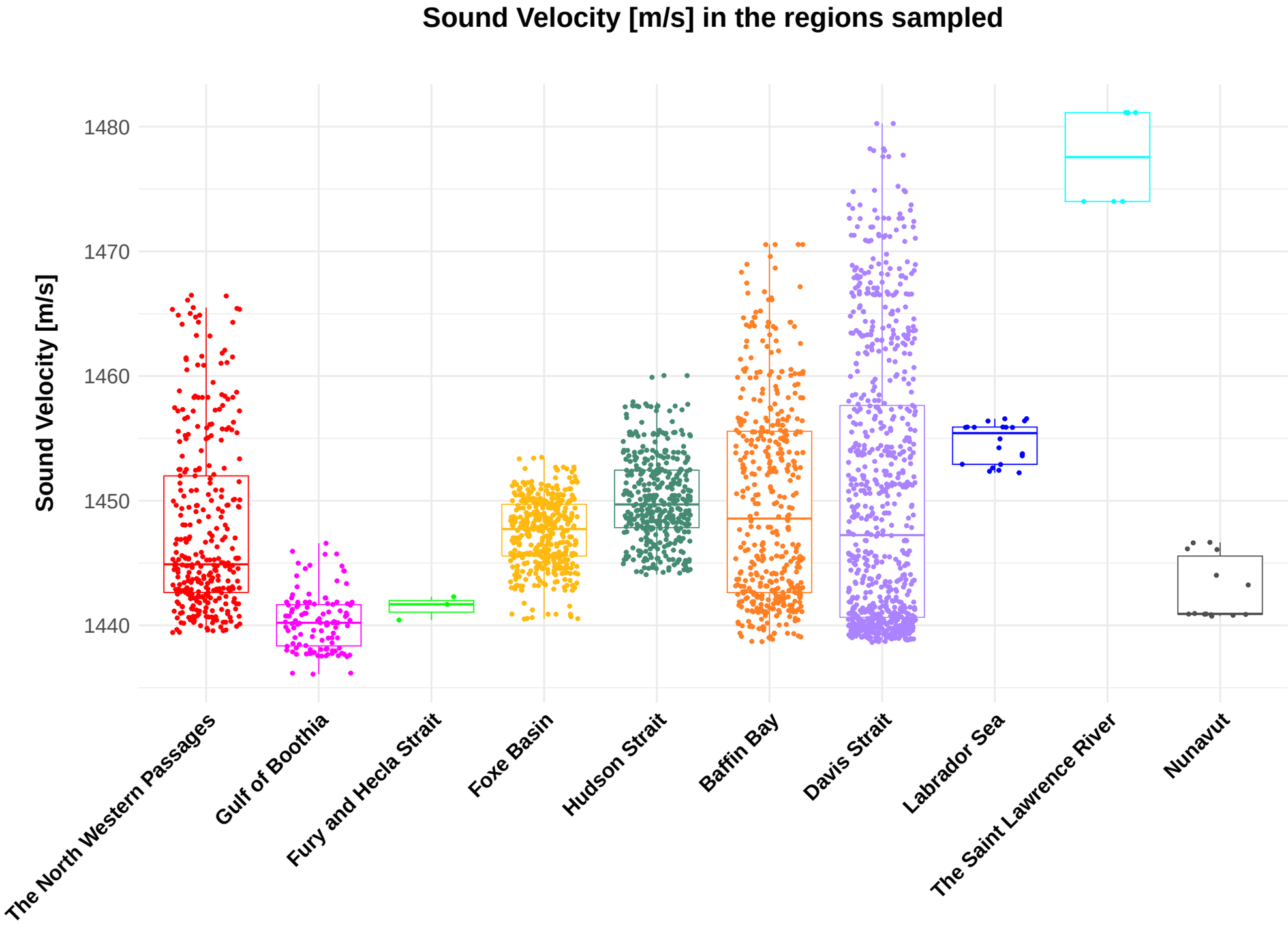

Longitude [deg] in the regions sampled
Longitude [deg]
−60
−70
−80
−90
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

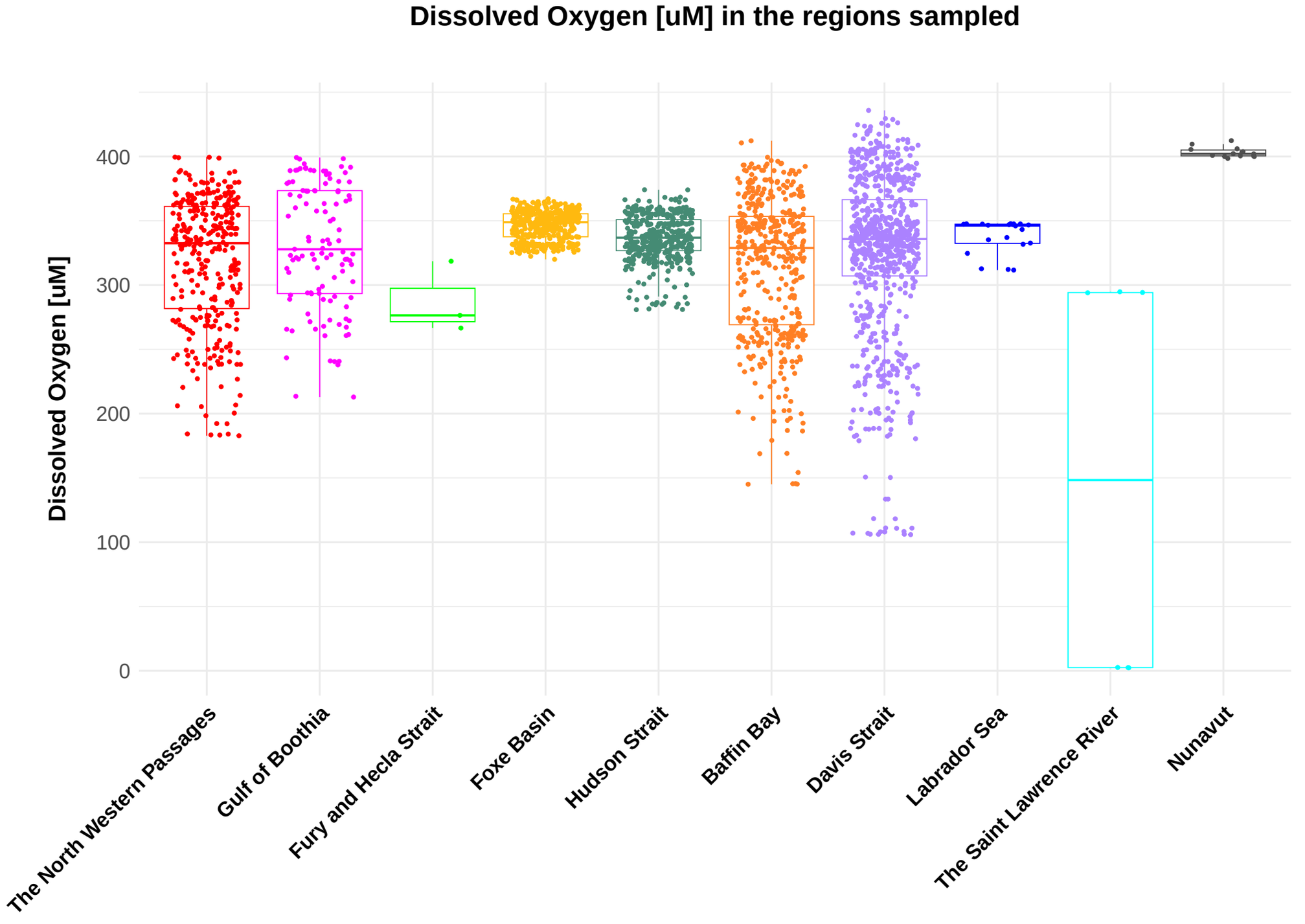

Temperature [deg C] in the regions sampled
Temperature [deg C]
9
6
3
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

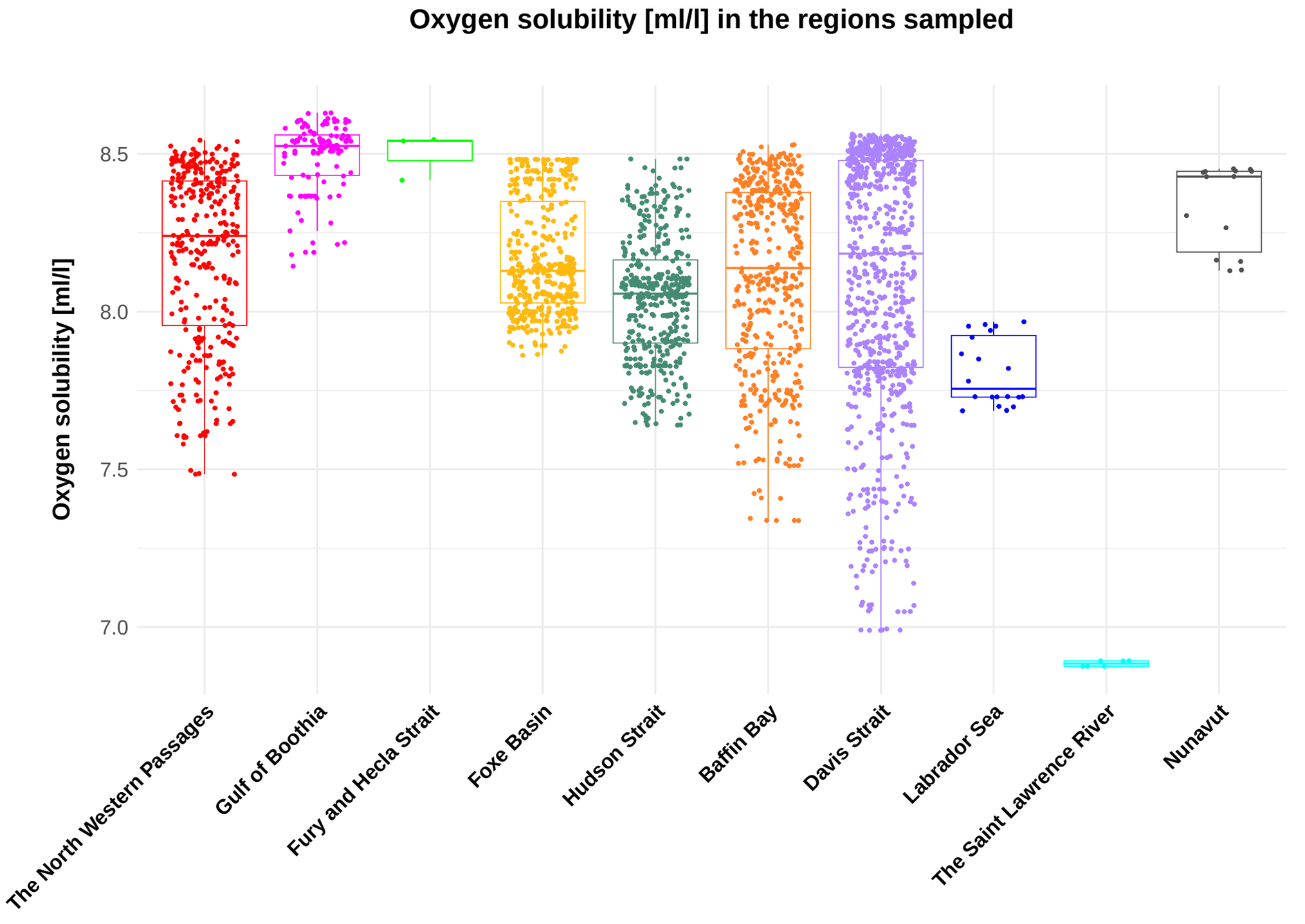

Salinity in the regions sampled
Salinity
35.0
32.5
30.0
27.5
25.0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

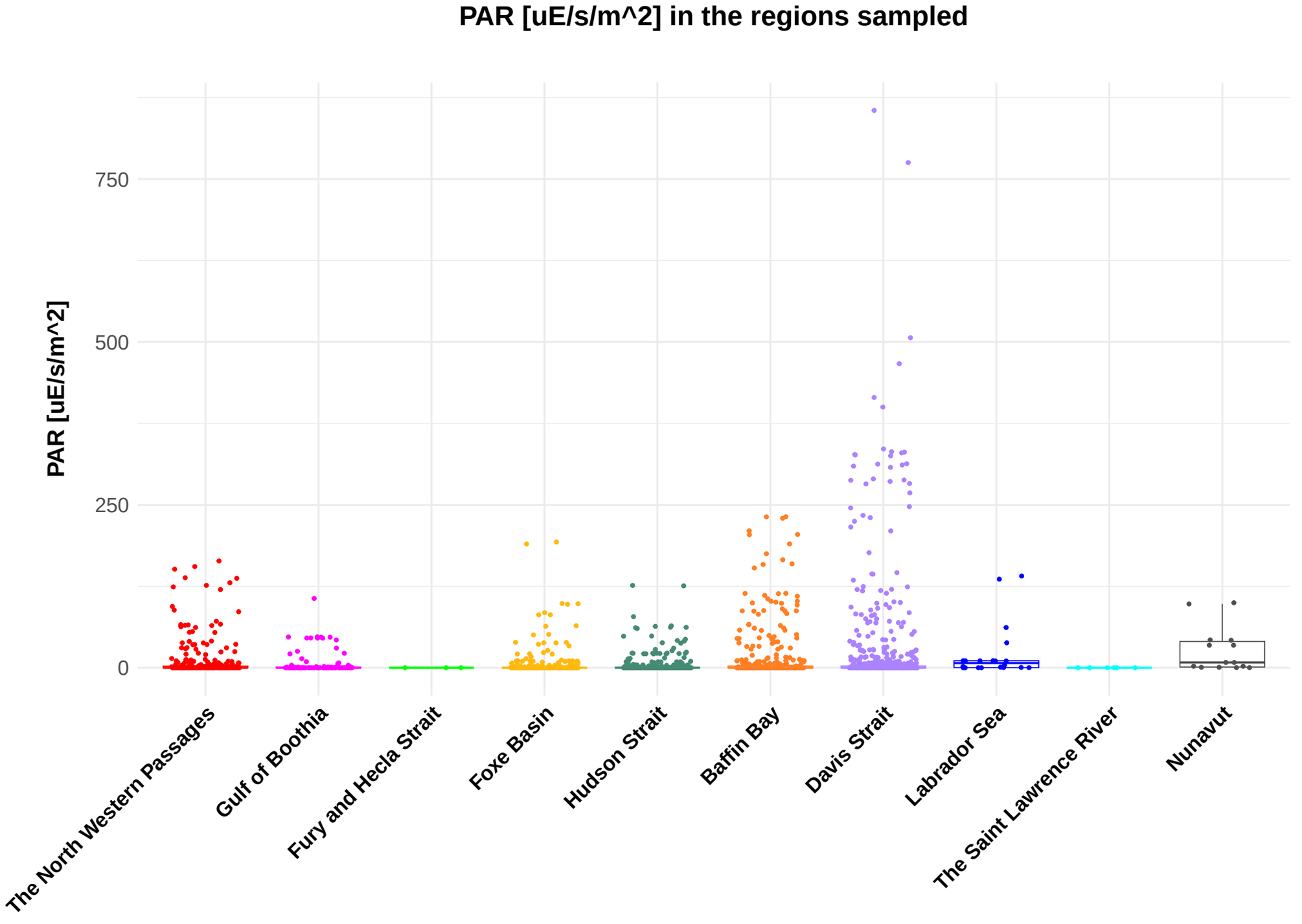

Sigma−T [kg/m^3] in the regions sampled
Sigma−T [kg/m^3]
28
26
24
22
20
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

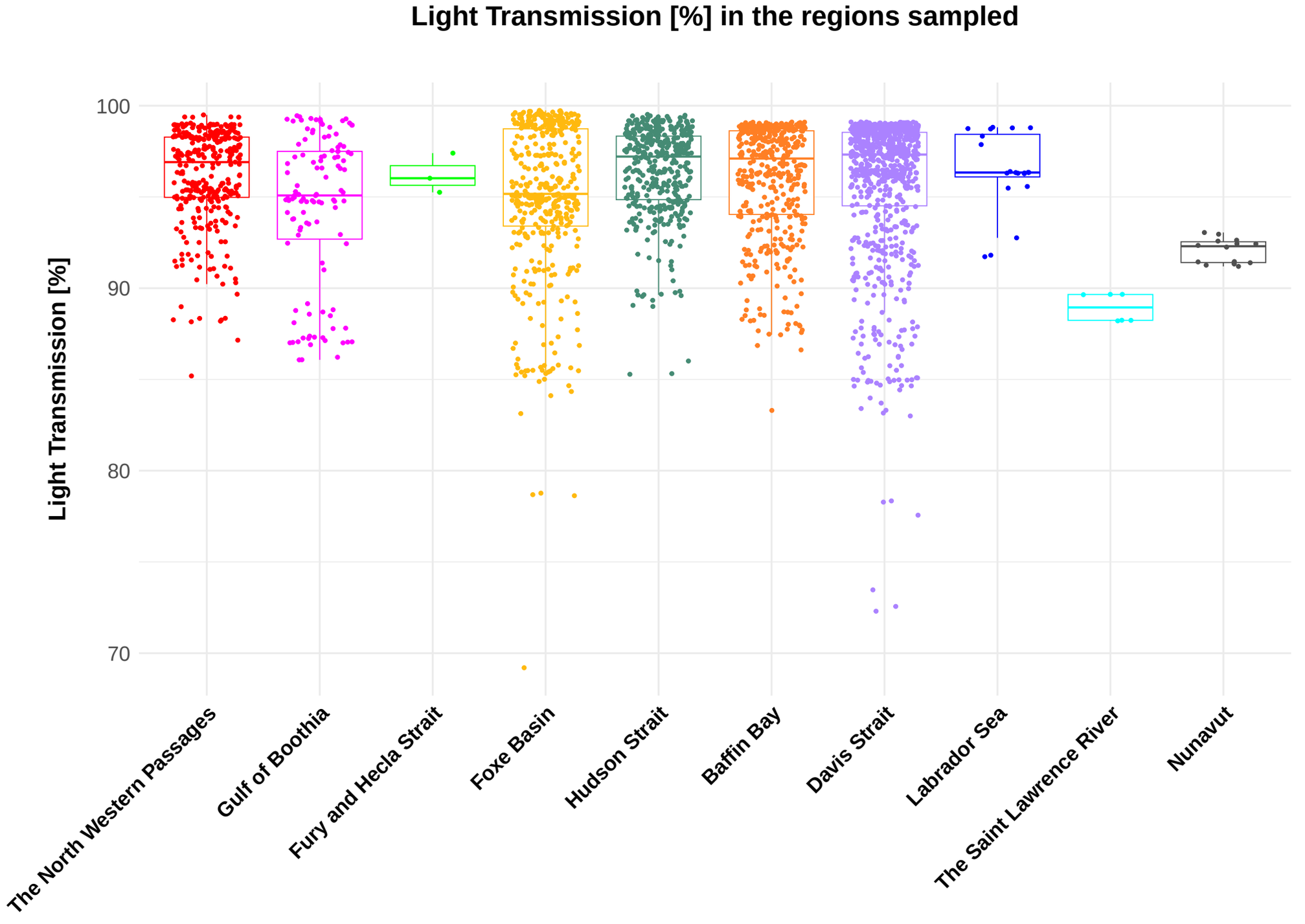

Sigma−Theta [kg/m^3] in the regions sampled
Sigma−Theta [kg/m^3]
28
26
24
22
20
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

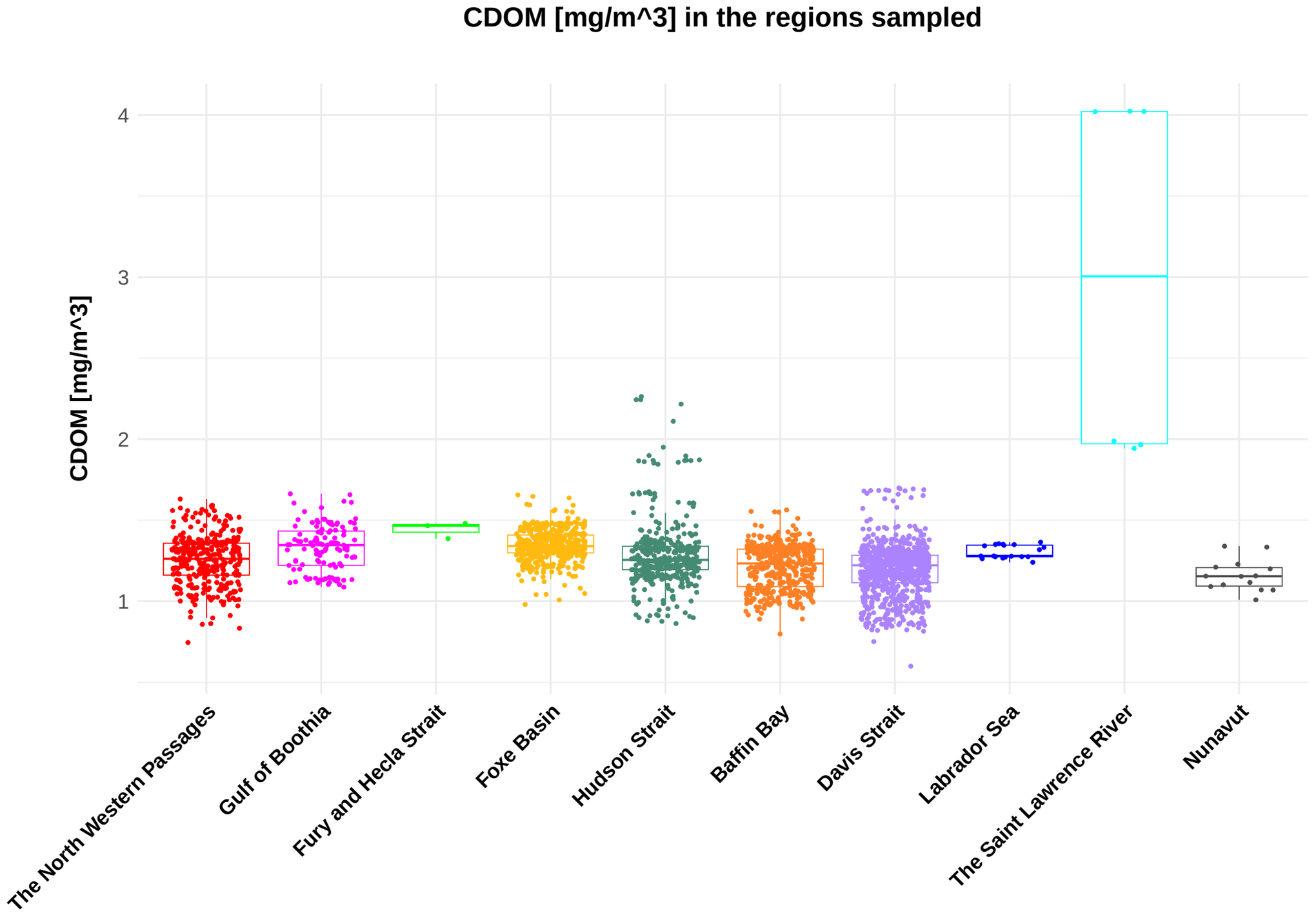

Sound Velocity [m/s] in the regions sampled
Sound Velocity [m/s]
1480
1470
1460
1450
1440
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

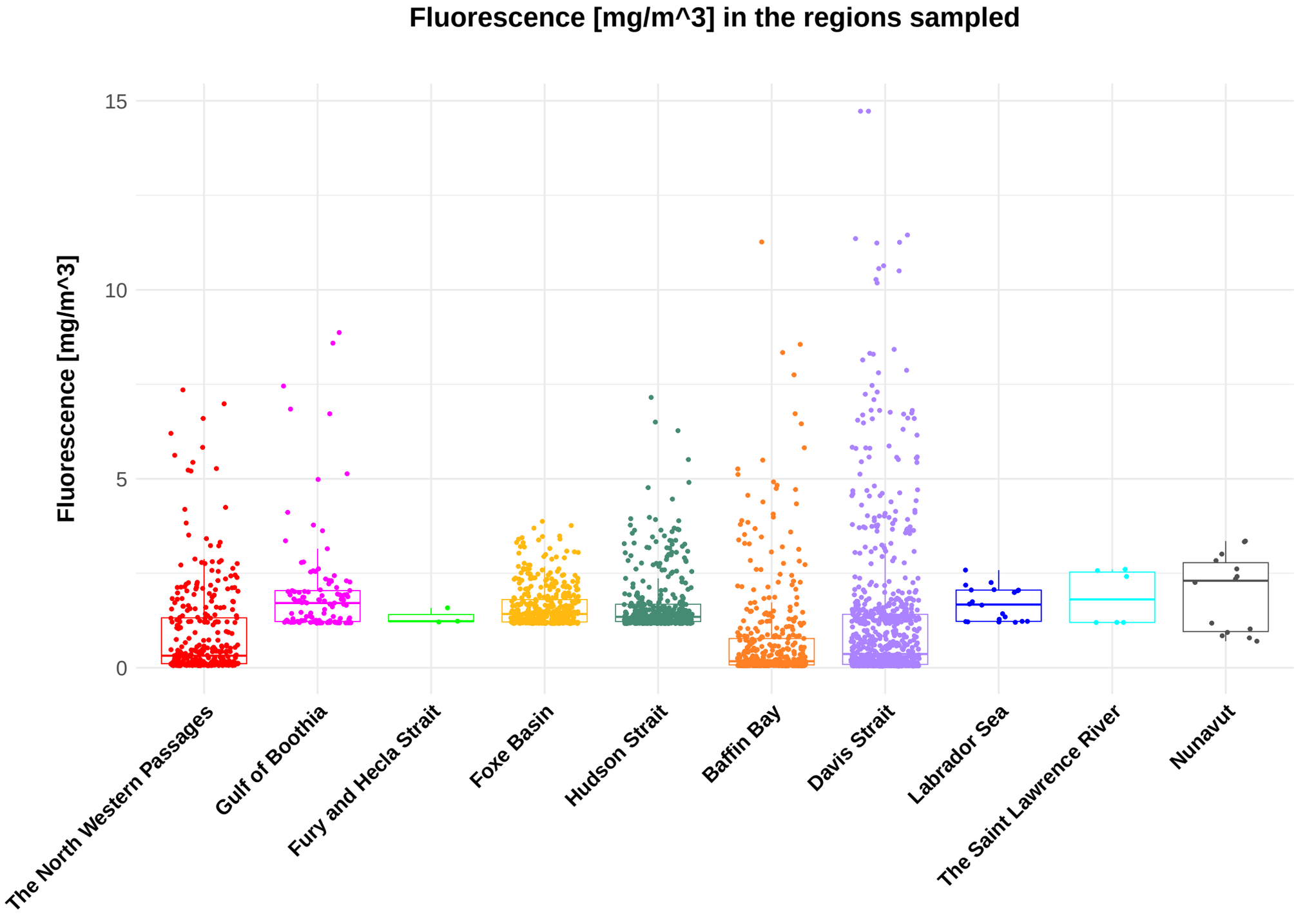

Dissolved Oxygen [uM] in the regions sampled
Dissolved Oxygen [uM]
400
300
200
100
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

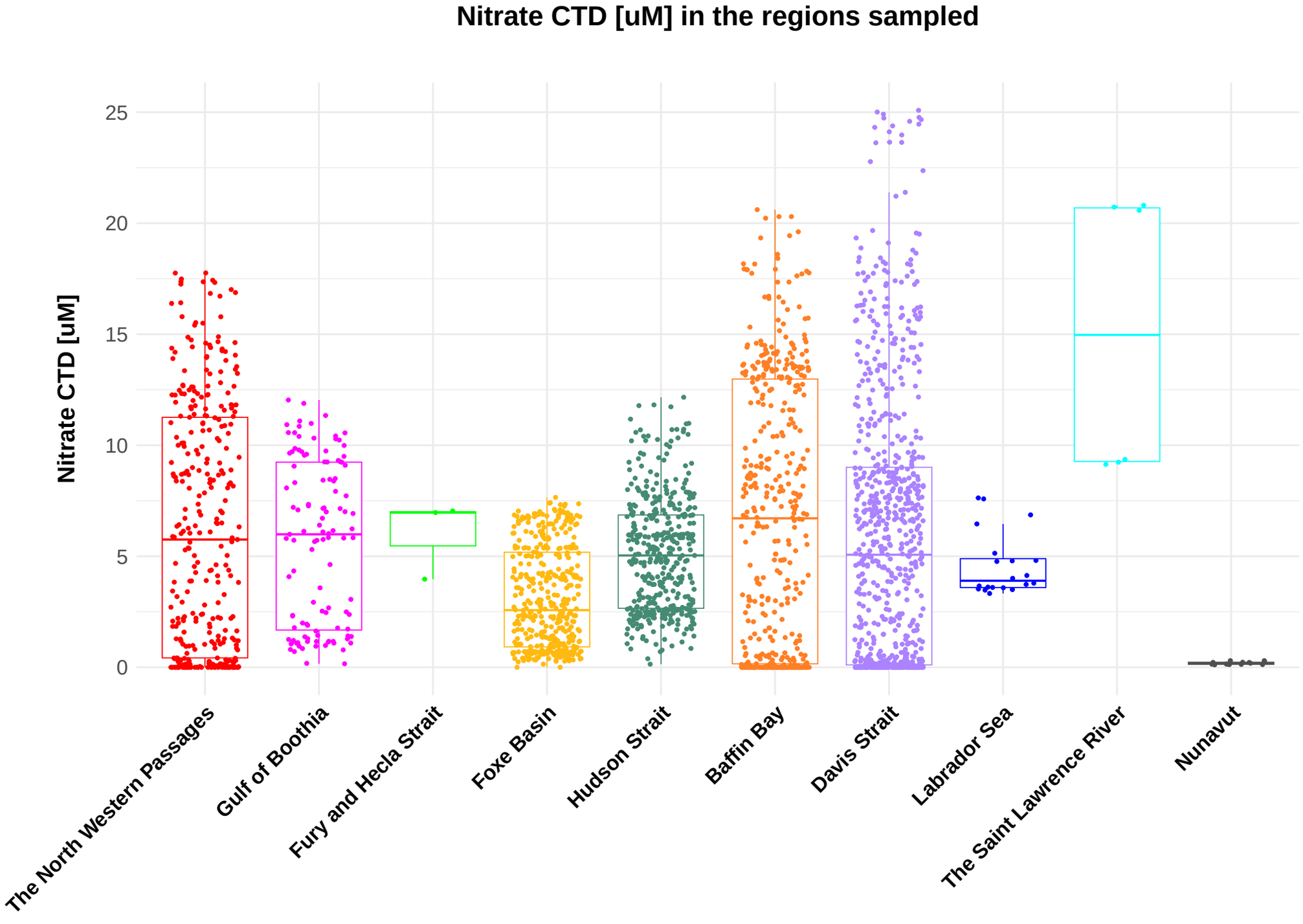

Oxygen solubility [ml/l] in the regions sampled
Oxygen solubility [ml/l]
8.5
8.0
7.5
7.0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

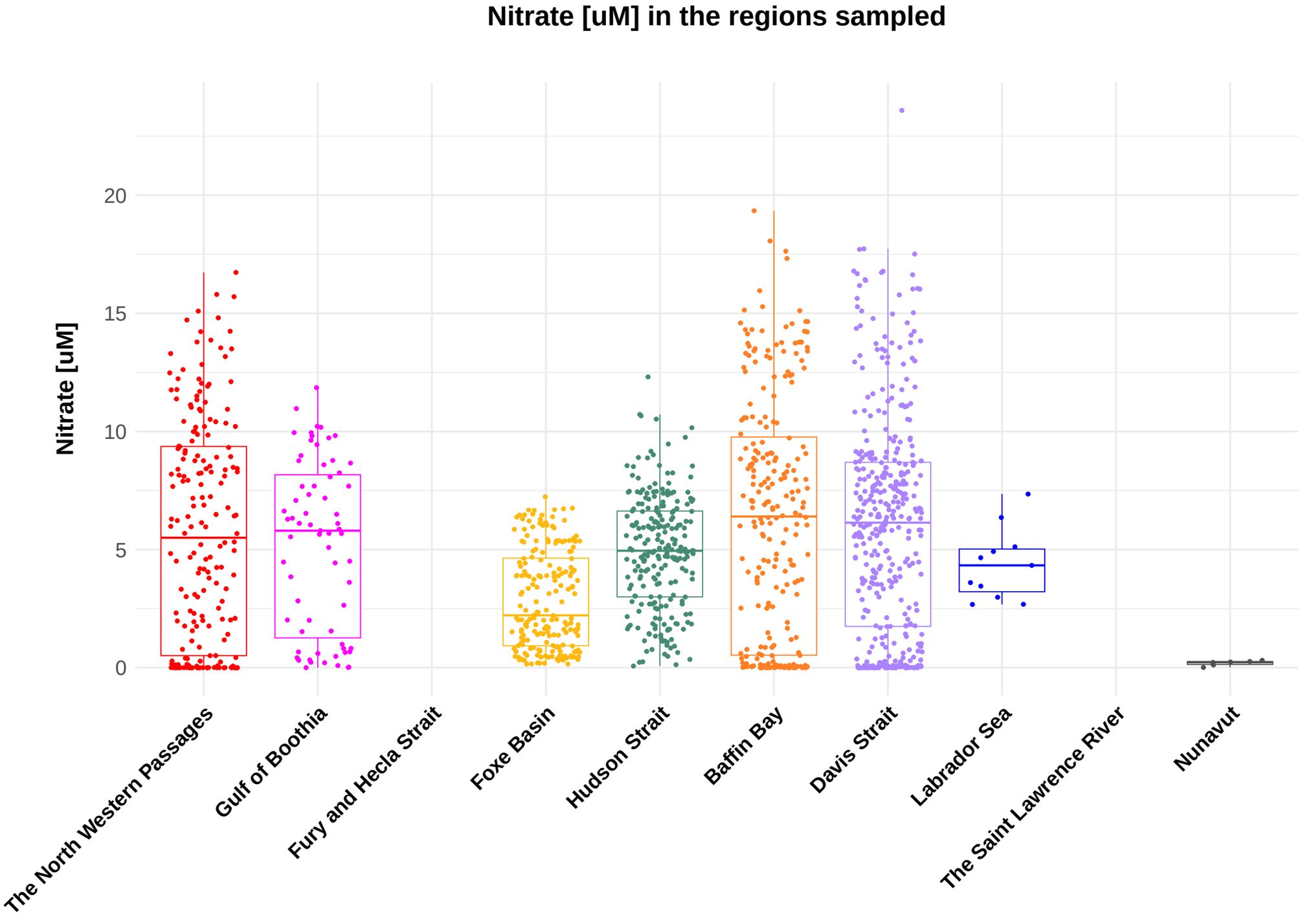

PAR [uE/s/m^2] in the regions sampled
PAR [uE/s/m^2]
750
500
250
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

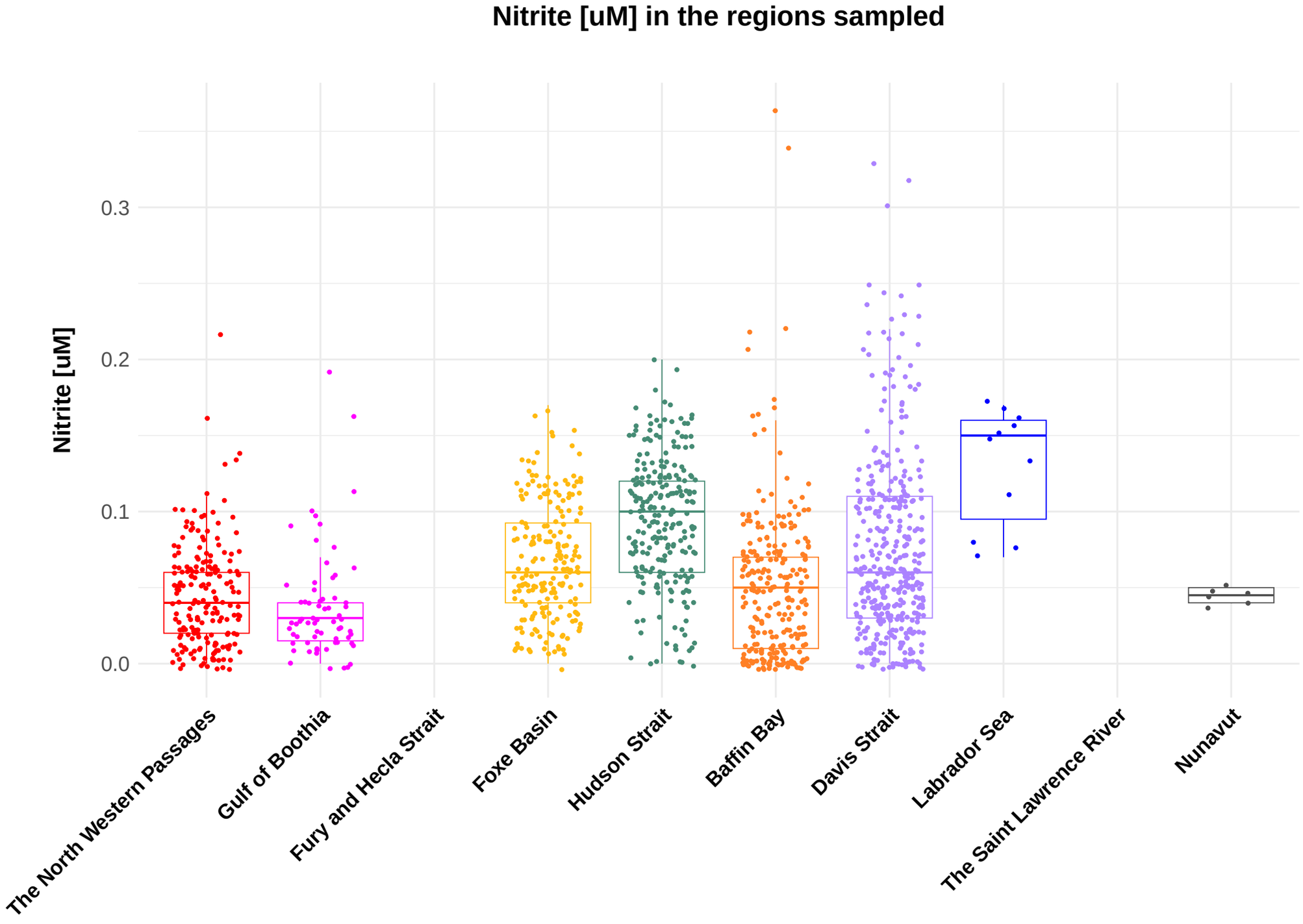

Light Transmission [%] in the regions sampled
Light Transmission [%]
100
90
80
70
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

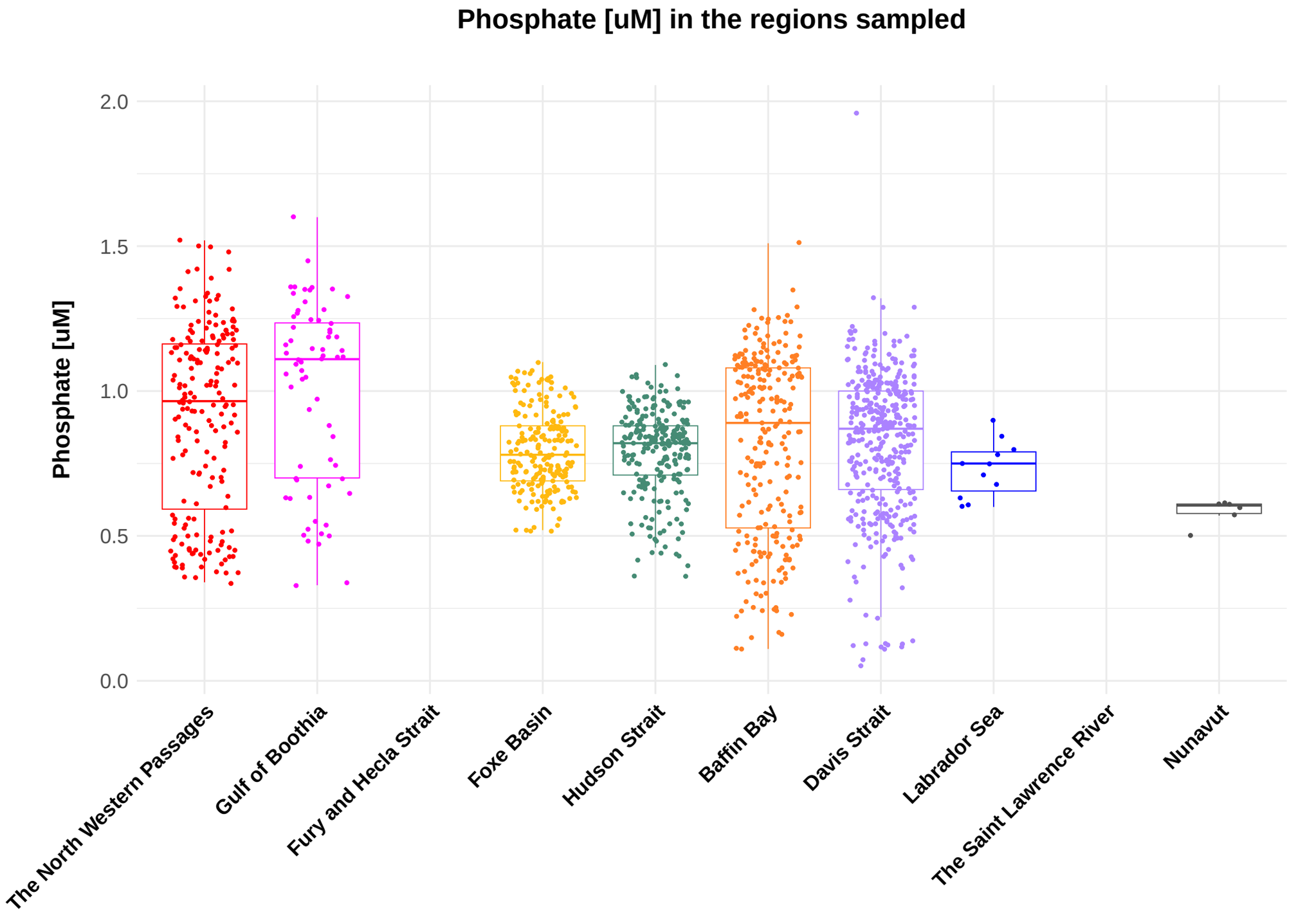

CDOM [mg/m^3] in the regions sampled
CDOM [mg/m^3]
4
3
2
1
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

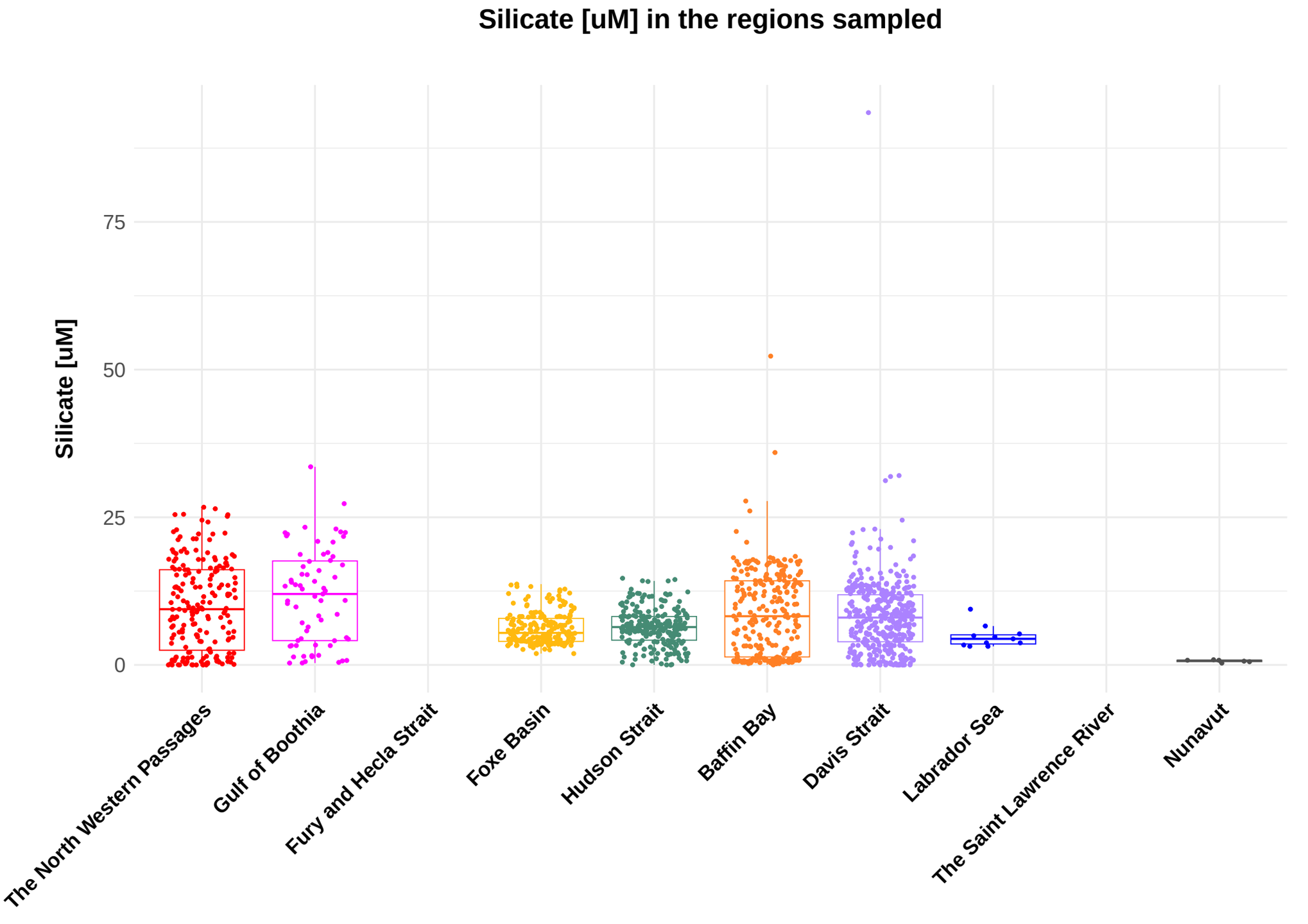

Fluorescence [mg/m^3] in the regions sampled
Fluorescence [mg/m^3]
15
10
5
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

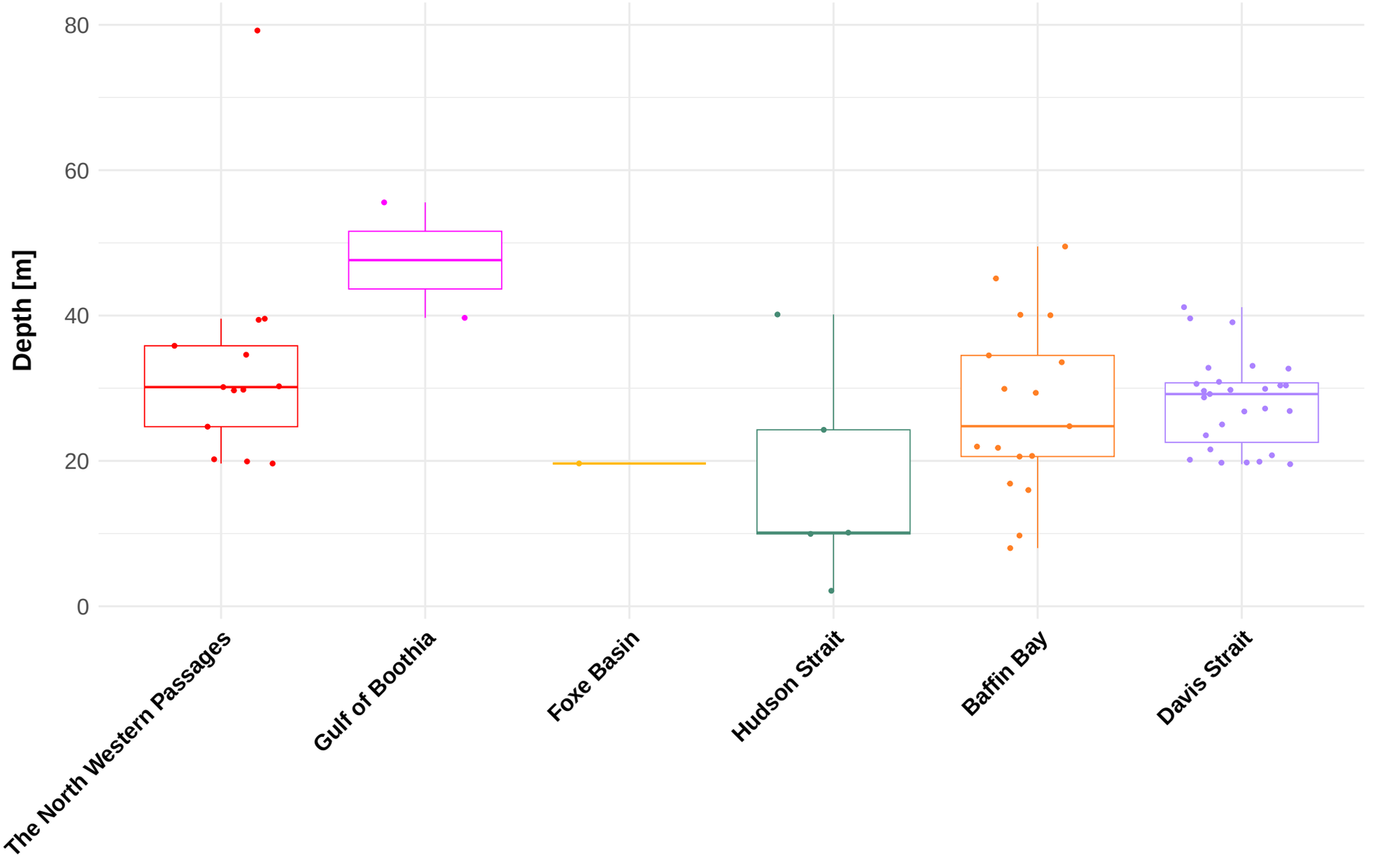

Nitrate CTD [uM] in the regions sampled
Nitrate CTD [uM]
25
20
15
10
5
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

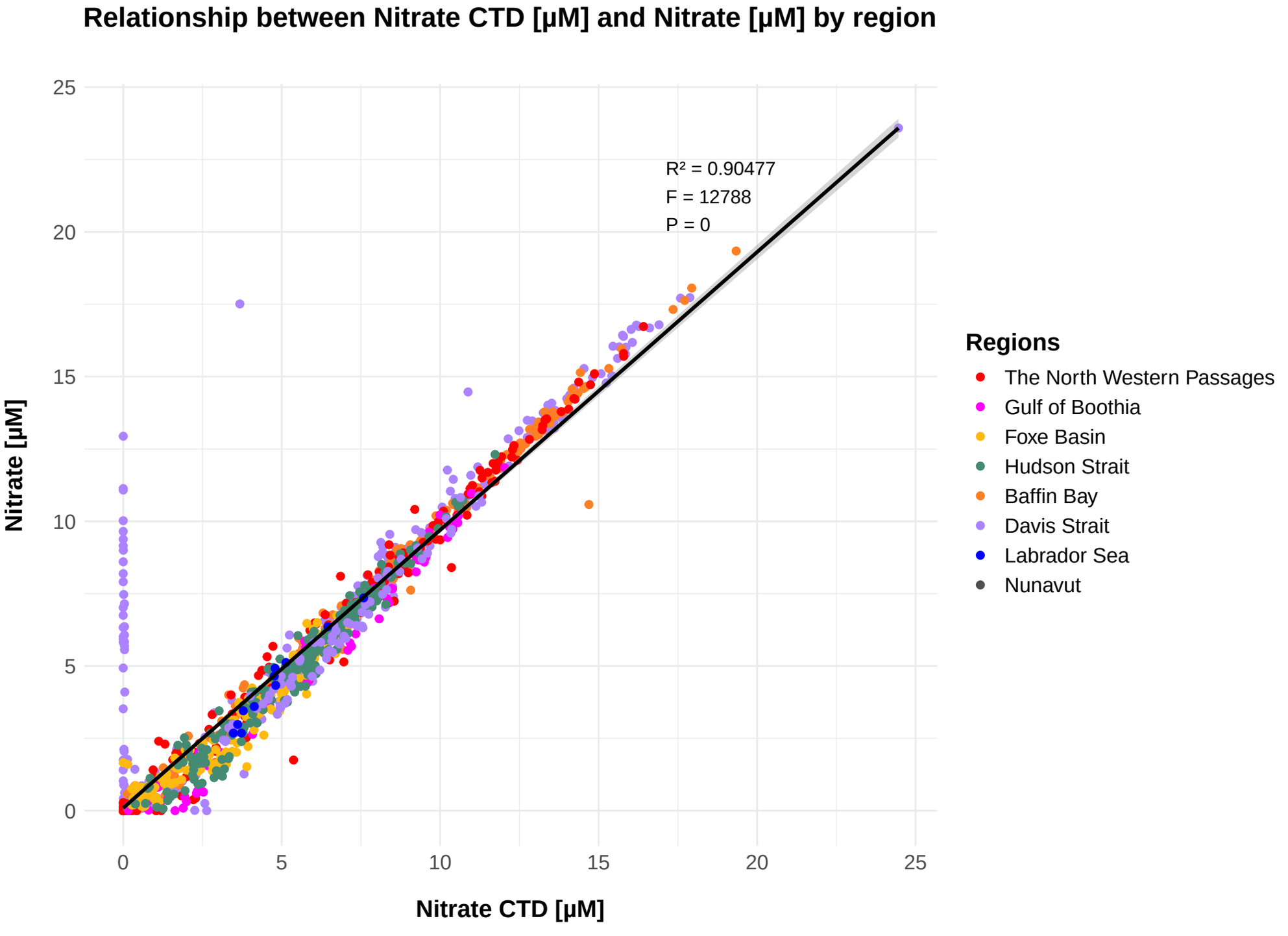

Nitrate [uM] in the regions sampled
Nitrate [uM]
0
5
10
15
20
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut

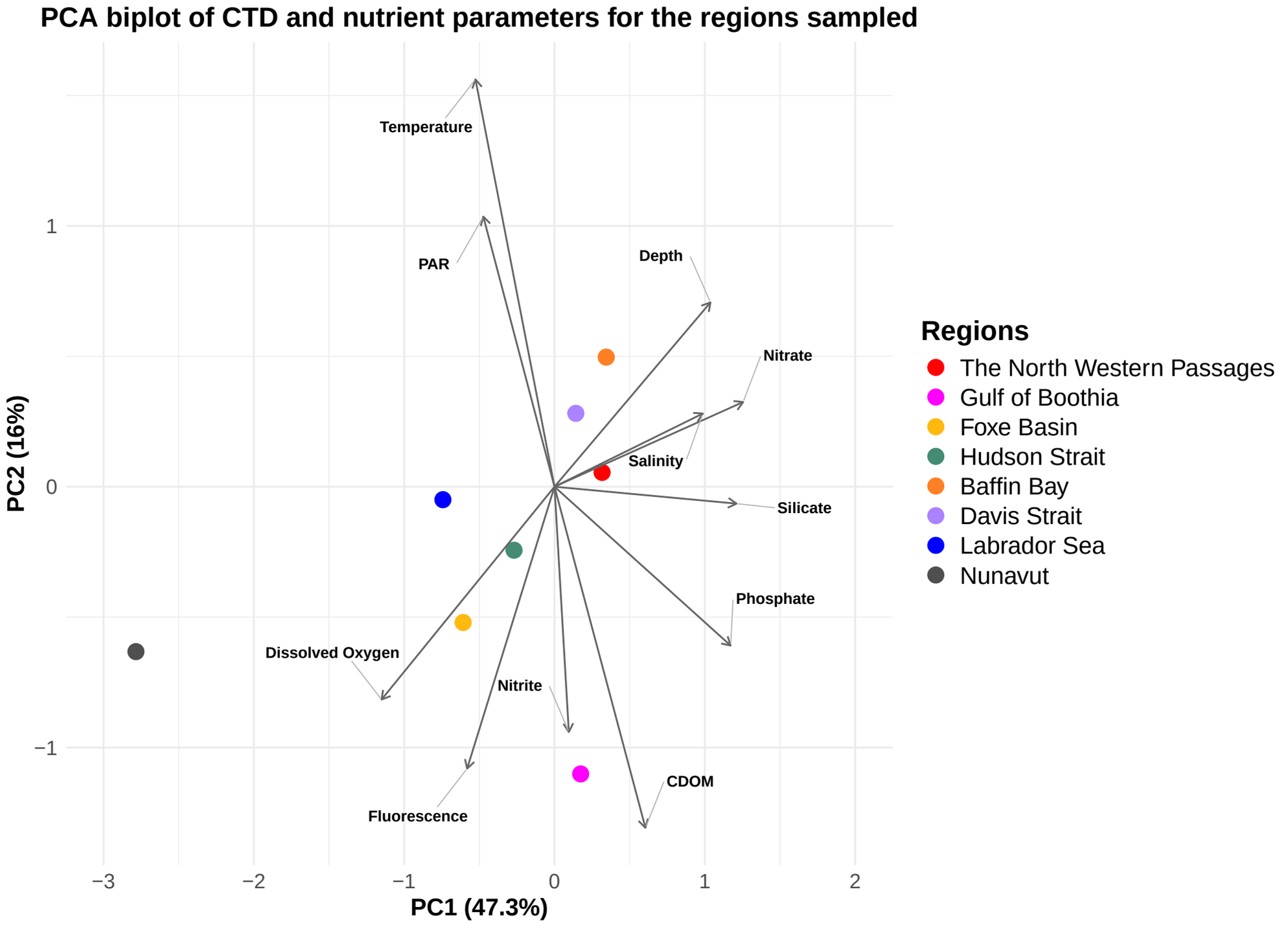

Nitrite [uM] in the regions sampled
Nitrite [uM]
0.0
0.1
0.2
0.3
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut


Phosphate [uM] in the regions sampled
Phosphate [uM]
2.0
1.5
1.0
0.5
0.0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut


Silicate [uM] in the regions sampled
Silicate [uM]
75
50
25
0
The North Western Passages
Gulf of Boothia
Fury and Hecla Strait
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
The Saint Lawrence River
Nunavut


Chl maximum depths in the regions sampled as inferred from fluorescence
(condition for fluorescence: max ≥ 2 × mean)
Depth [m]
80
60
40
20
0
The North Western Passages
Gulf of Boothia
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait


Relationship between Nitrate CTD [µM] and Nitrate [µM] by region
R² = 0.90477
F = 12788
P = 0
Nitrate [µM]
Nitrate CTD [µM]
0
5
10
15
20
25
Regions
The North Western Passages
Gulf of Boothia
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
Nunavut


PCA biplot of CTD and nutrient parameters for the regions sampled
Temperature
PAR
Depth
Nitrate
Salinity
Silicate
Phosphate
Dissolved Oxygen
Nitrite
CDOM
Fluorescence
PC2 (16%)
PC1 (47.3%)
1
0
−1
−3
−2
−1
0
1
2
Regions
The North Western Passages
Gulf of Boothia
Foxe Basin
Hudson Strait
Baffin Bay
Davis Strait
Labrador Sea
Nunavut